lecture 17 Newton's Method and Nonlinear Equation Solving @April 5, 2026

Summary

Executive Summary

- Newton's method solves nonlinear equations by iteratively linearizing the function and finding where the linear approximation crosses the x-axis, yielding quadratic convergence for simple roots when starting close enough.▸▸
- The method requires computing derivatives and may fail to converge if the initial guess is too far from the root or if the derivative vanishes (e.g., at double roots).▸▸
- The secant method approximates the derivative using function values from previous iterations, eliminating the need for explicit derivative calculations while maintaining superlinear convergence with rate $\phi \approx 1.618$ (golden ratio).▸▸
- Fixed point iteration reformulates $f(x) = 0$ as $x = g(x)$ and converges linearly when g is a contraction (i.e., $|g'(x)| \leq \rho < 1$); the art lies in choosing an appropriate g.▸▸
- Newton's method extends naturally to systems of n nonlinear equations in n unknowns by replacing scalar derivatives with the Jacobian matrix, requiring solution of a linear system at each iteration.▸▸
- Exam-relevant: Understand convergence requirements (simple root, close initial guess), recognize failure modes (double roots, poor initialization), and be able to implement and analyze Newton's method for both single equations and systems.▸▸

Conceptual Map

Newton's Method for Scalar Equations

(A) Core Concept & Context

Newton's method solves $f(x) = 0$ by constructing a first-order Taylor approximation $\hat{f}$ at the current iterate $x^{(k)}$ and setting it to zero to obtain the next iterate $x^{(k+1)}$.▸▸ This addresses the fundamental problem that nonlinear equations lack closed-form solutions, enabling iterative approximation with rapid (quadratic) convergence under suitable conditions.▸

(B) Detailed Explanation

At iteration k, linearize f around $x^{(k)}$:

$$f(x) \approx \hat{f}(x) = f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)})$$

Setting $\hat{f}(x) = 0$ and solving for x yields the update formula.▸ Graphically, this corresponds to drawing the tangent line to f at $x^{(k)}$ and finding where it intersects the x-axis; that intersection point becomes $x^{(k+1)}$.▸▸

Assumptions and failure modes:

- f'' must be continuous in the interval of interest.▸
- The root must be simple (i.e., $f'(r) \neq 0$).▸▸
- Initial guess must be sufficiently close to the root; otherwise the method may diverge or converge to a different root.▸▸
- Method fails when $f'(x^{(k)}) \approx 0$, especially at double roots where both $f(r) = 0$ and $f'(r) = 0$.▸▸

(C) Formal Notations (LaTeX)

∑ **Newton's update rule**

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$$

Definitions:

- $x^{(k)}$: current iterate
- $x^{(k+1)}$: next iterate
- $f(x^{(k)})$: function value at current iterate
- $f'(x^{(k)})$: derivative at current iterate (must be nonzero)

Taylor truncation error:

$f(x) - \hat{f}(x) = \frac{1}{2}(x-z)^2 f''(\xi)$

where ξ is a point close to z.▸

Quadratic convergence:

$|e_{k+1}| \leq M_1 |e_k|^2$

where $e_k = x^{(k)} - r$ and M_1 is a positive constant.▸

(D) Critical Connections

- Taylor series expansion is the mathematical foundation; Newton's method uses only the first-order term.▸▸
- Bisection method guarantees convergence but is slow (linear rate); Newton trades robustness for speed.▸
- For double roots (e.g., $f(x) = x^2$ at $x = 0$), convergence degrades from quadratic to linear because f' is small near the root.▸▸
- The method is a special case of fixed point iteration with $g(x) = x - f(x)/f'(x)$.▸

(E) Evidence from the transcript

- "You linearize the function about one point, and in this case it's the previous iteration, right? And then once you have that approximation, you set it to zero."▸
- "For mutants to converge... you have to be really close to the root."▸
- Example shown: starting at $x_0 = 0.9$ converges, but starting at $x_0 = 1.1$ diverges for the same function.▸▸

Secant Method (e.g. Quasi-Newton)

(A) Core Concept & Context

The secant method approximates $f'(x^{(k)})$ using a finite difference of previous function evaluations, eliminating the need to compute derivatives analytically or via automatic differentiation.▸ This resolves the computational cost issue when derivatives are expensive or unavailable.▸

(B) Detailed Explanation

Replace the derivative in Newton's formula with:

$f'(x^{(k)}) \approx \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$

The method constructs a secant line (passing through two points on the graph) rather than a tangent line, then finds where the secant crosses the x-axis.▸▸ This requires two initial guesses instead of one.▸▸

Boundary condition: The approximation is valid when successive iterates are distinct and function evaluations are available from the iteration history.▸

(C) Formal Notations (LaTeX)

∑ **Secant update rule**
$x^{(k+1)} = x^{(k)} - f(x^{(k)}) \cdot \frac{x^{(k)} - x^{(k-1)}}{f(x^{(k)}) - f(x^{(k-1)})}$

Definitions:

- Requires two starting points: $x^{(0)}$ and $x^{(1)}$
- Uses function values from current and previous iterations
- No derivative evaluation required

Convergence rate:

$r = \phi = \frac{1+\sqrt{5}}{2} \approx 1.618$

The golden ratio; superlinear but subquadratic.▸▸

(D) Critical Connections

- Secant is faster than bisection (linear convergence) but slower than Newton's (quadratic convergence).▸▸
- The finite difference approximation to f' provides "more information than bisection, but not as much as Newton's," resulting in intermediate convergence speed.▸
- Like Newton's, the secant method requires a sufficiently close initial guess and offers no convergence guarantee.▸▸
- Implementations may drop either the left or right endpoint at each iteration; some use smart strategies to decide.▸▸

(E) Evidence from the transcript

- "Instead of working with f', I work with an approximation to f'... I already have the f's from the previous iterations."▸▸

- "The rate of convergence is the golden ratio, so 1.6. So it's super linear... but it's not as good as quadratic as the Newton's method."▸

Fixed Point Iteration

(A) Core Concept & Context

A fixed point of g satisfies $x^ = g(x^)$, i.e., the function value equals the input.▸ Fixed point iteration reformulates the root-finding problem $f(x) = 0$ as $x = g(x)$ by constructing a suitable g, then iterates $x^{(k+1)} = g(x^{(k)})$ until convergence.▸▸ This provides a unifying framework for many iterative methods, including Newton's and secant as special cases.▸▸

(B) Detailed Explanation

Graphical interpretation: Plot $y = g(x)$ and $y = x$; fixed points occur at intersections.▸ The iteration produces a "cobweb" diagram: starting from $x^{(0)}$, move vertically to $(x^{(0)}, g(x^{(0)}))$, then horizontally to the line $y = x$ at $(x^{(1)}, x^{(1)})$, and repeat.▸▸

Reformulation: Given $f(x) = 0$, construct g such as:

- $g(x) = x - f(x)$▸
- $g(x) = x + 2f(x)$▸
- $g(x) = x - f(x)/f'(x)$ (Newton's method)▸

The art lies in choosing g with appropriate convergence properties.▸

Existence and uniqueness conditions:

1. **Existence:** If g is continuous on $[a, b]$ with $g(a) \geq a$ and $g(b) leq b$, then a fixed point exists in $[a, b]$.▸▸
2. **Uniqueness:** If $|g'(x)| \leq \rho < 1$ for all $x in [a, b]$, then g is a **contraction** and the fixed point is unique.▸

Failure mode: If $g'(x) > 1$ in the region of interest, the iteration diverges.▸▸

(C) Formal Notations (LaTeX)

∑ **Fixed point iteration**
$x^{(k+1)} = g(x^{(k)})$
Contraction condition
$|g'(x)| \leq \rho < 1 \quad \forall x \in [a, b]$
Error bound
$|x^{(k+1)} - x^*| \leq \rho |x^{(k)} - x^*|$
Iterating from $k = 0$ to k:
$|x^{(k)} - x^*| \leq \rho^k |x^{(0)} - x^*|$
Since $rho < 1$, $\rho^k \to 0$ as $k to infty$, guaranteeing convergence.▸

Definitions:

- x^*: fixed point
- g: contraction mapping
- rho: Lipschitz constant, $\rho < 1$
- Convergence is **linear** with rate rho▸

(D) Critical Connections

- Proof of existence uses the intermediate value theorem.▸
- Convergence analysis parallels bisection method but with rate ρ instead of $1/2$; smaller ρ yields faster convergence.▸▸
- Newton's method is a fixed point iteration with $g(x) = x - f(x)/f'(x)$, which is a contraction near simple roots.▸
- The power method for eigenvalues is also an iterative method, conceptually related to fixed point iteration.▸▸

(E) Evidence from the transcript

- "You draw the graph of your function, all the points that intersect the line $y = x$, those are the fixed points."▸
- "Not all g's are good... some g's are good and some g's are not good... The first is you have to make sure that the g has a fixed point... and the second part... has to do with uniqueness."▸▸
- Cobweb diagram construction explained in detail.▸▸

Newton's Method for Systems of Nonlinear Equations

(A) Core Concept & Context

The method extends to solve $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ where $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$ by linearizing each component f_i and solving the resulting linear system at each iteration.▸▸ This addresses the critical need to solve coupled nonlinear PDEs, optimization problems, and boundary value problems where multiple equations must be satisfied simultaneously.▸

(B) Detailed Explanation

Construct the function $\mathbf{f} = (f_1, f_2, \ldots, f_n)^T$ where each f_i corresponds to one equation.▸▸ The linear approximation of each component is:

$$\hat{f}_i(x) = f_i(\mathbf{z}) + \nabla f_i(\mathbf{z})^T(\mathbf{x} - \mathbf{z})$$

where ∇f_i is the gradient (row vector) of f_i.▸

Stack these to form:

$$\hat{\mathbf{f}}(\mathbf{x}) = \mathbf{f}(\mathbf{z}) + J(\mathbf{z})(\mathbf{x} - \mathbf{z})$$

where J is the **Jacobian matrix** with entries $J_{ij} = partial f_i / partial x_j$.▸▸

Setting $\hat{\mathbf{f}}(\mathbf{x}) = \mathbf{0}$ and choosing $\mathbf{z} = \mathbf{x}^{(k)}$ yields:

$$J(\mathbf{x}^{(k)})(\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}) = -\mathbf{f}(\mathbf{x}^{(k)})$$

Algorithm:

1. Compute Jacobian $J(\mathbf{x}^{(k)})$
2. Solve linear system $J(\mathbf{x}^{(k)})\Delta\mathbf{x} = -\mathbf{f}(\mathbf{x}^{(k)})$ for $\Delta\mathbf{x}$
3. Update: $mathbf{x}^{(k+1)} = mathbf{x}^{(k)} + Delta mathbf{x}$▸▸

Critical: Never explicitly invert the Jacobian; solve the linear system using LU, QR, or other factorizations.▸▸

(C) Formal Notations (LaTeX)

$\sum$ **Jacobian matrix**

$$J(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{pmatrix}$$

Newton iteration (system form)

$$J(\mathbf{x}^{(k)})\Delta\mathbf{x} = -\mathbf{f}(\mathbf{x}^{(k)})$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta\mathbf{x}$$

Definitions:

- $mathbf{f}: mathbb{R}^n to mathbb{R}^n$: vector-valued function
- $J(mathbf{x})$: $n \times n$ Jacobian matrix
- $Delta mathbf{x}$: update vector
- Row i of J is $nabla f_i$▸

Quadratic convergence (systems):

$$\|\mathbf{e}_{k+1}\| \leq M_1\|\mathbf{e}_k\|^2$$

$$\|\mathbf{f}(\mathbf{x}^{(k+1)})\| \leq M_2\|\mathbf{f}(\mathbf{x}^{(k)})\|^2$$

where $\|\cdot\|$ denotes a vector norm.▸▸

(D) Critical Connections

- Conceptually identical to scalar Newton's: linearize, solve for zero of the approximation, iterate.▸
- Each iteration requires solving an $n \times n$ linear system, but fast solvers make this tractable even for large n.▸
- Example application: discretized nonlinear ODE/PDE yields a system of nonlinear equations (e.g., $u'' = e^u$ with boundary conditions).▸▸
- Eigenvalue computation via characteristic polynomial $\det(A - \lambda I) = 0$ is another nonlinear equation; solving it with Newton's finds all eigenvalues simultaneously.▸▸

(E) Evidence from the transcript

- "It's the same process, only the calculation looks a little bit busier."▸

- "You don't want to calculate this guy [the inverse]. What you do is you calculate the Jacobian at iteration $x^{(k)}$... and then you set it to the negative f at $x^{(k)}$... you solve for delta x."▸
- Example with nonlinear BVP $u'' = e^u$, $u(0) = u(1) = 0$, discretized into $n-1$ equations.▸▸

Comparisons and supporting details

Method
Bisection
Newton's
Sec
Fixed Point

Computational cost considerations:

- Newton's requires derivative evaluation at each step, which may be expensive.▸▸
- Systems of nonlinear equations require solving a linear system per iteration (manageable with modern solvers).▸
- Automatic differentiation can compute derivatives but adds computational overhead.▸

Convergence diagnostics:

- Plot residual $|f(x^{(k)})|$ vs. iteration in semi-log scale; quadratic convergence appears as accelerating descent, linear as a straight line.▸▸
- For systems, use $\|\mathbf{f}(\mathbf{x}^{(k)})\|$ as the convergence metric.▸▸
- Parabola-like appearance in semi-log plots does not guarantee quadratic convergence; verify the rate analytically.▸

Number of iterations to reduce error by factor of 10:

For fixed point iteration with linear rate rho:

$$\frac{|e_k|}{|e_0|} = 0.1 \implies \rho^k = 0.1 \implies k = \frac{\log(0.1)}{\log(\rho)} = \frac{-1}{\log(\rho)}$$

▸▸

Action Items

- [ ] Review homework problems on Newton's method convergence for double roots (linear vs. quadratic rates)▸
- [ ] Practice timing yourself on sample problems for exam preparation▸
- [ ] Study the list of topics provided for the exam (90-minute duration)▸▸

Notes

Transcript

Newton's method is based on a linear approximation of your function. So in the blue you have the function or the graph of the function f of x. And f hat is the linear approximation. So it's the dashed line that you see there. So the dashed red line is actually the graph of the best linear approximation to the function f. Okay, and this means that the tangent line to the graph of the function, so the derivative of f hat at c is the same as the derivative of f.

So the idea is, or the background to the Newton's method is Taylor approximation. So I have there in the first item, I am getting a linear approximation of my function f in a neighborhood of z. Okay, and that linear approximation is what I'm calling f-hat. So it's a linear function in the x. So remember z is known. So this is basically Taylor's up to the first order. So the next term would be a term of the second order.

So we say that the error, and this is the truncation error, can be calculated as, so this is the truncation error, okay, between f and the linear approximation, is one half of x minus e squared multiplied, so the truncation error would be multiplied by the second derivative, by the second derivative in the close by, evaluating the close point that I'm holding aside. And then what I've done here is I calculated an upper bound for the second order derivative.

And this is the value between-or the error between the function itself And, move your feet.

So this is Taylor series, something that you've seen so far before. So the idea is that instead of working with f of x, So instead of working with the function with the blue graph, what I'm going to do is I'm going to be working with the approximation. or in the neighborhood of x equals 2z. Okay, so this is the idea behind new tips. And... And Following up on mutants, I'm going to replace my function by its linear approximation f hash.

And this is done at the iteration k, so at x of k. So the idea is I'm going to set my z. to be x of k, so the kth iterate.

So this C oops So this guy is going to be x. Okay, right? So I'm going to take the, I'm going to write my z to be x power k plus 1 like this. So this is also my z, which is going to go, not xk plus 1. This is what I'm going to write after. iteration x k so my z is iteration f k so my z becomes the current iteration, so I have to calculate the F at my current iteration. I have to calculate the derivative at the current iteration, and then I just have my F hat here.

here as a function of x, so the point that you are considering in the neighborhood of the previous iteration, so x of k.

And then...

What you want to do, you want to make this equal to 0.

So this is the equation that you want to solve. So your only unknown is this x.

And you believe that this x is going to be the next iteration. So you call it x of k plus 1. So when you solve this equation, which is the same as this one, in terms of my x, I can find that my x, which is my next iteration, is going to be just the previous iteration with this update here. So the update that you see here...

It depends on the value of the function at the previous iterate and the value of the derivative of the function And this update, of course, is only valid assuming that the first derivative in the previous iteration, so my x of k, is different from 0.

And this is Newton's method.

So Newton's method, so this is like the analytic part of Newton's. In terms of graphical visualization of what's going on.

It's very simple. So you are at x of k, right? You do, so this is the graph of the function that you are working because you are not, When you're working with newtons, you will not work with the function, but it's linear approximation that you use from Taylor's, okay? So, F. X iteration K, you linearize the function, or I mean, you linearize the function so you obtain the linear approximation that you are calling F hat.

And then what you do is This F hat is set to 0 so that you get to this approximation, which means that you are looking for the point along the x-axis where the linear approximation hits the 0. And this would be the value of the next iteration.

So this is how newtons work. So I've presented, so everything is based on the linear approximation of your function.

You linearize the function about one point, and in this case it's the previous iteration, right? And then once you have that approximation, you set it to zero.

and then you find the value you sold for the value x. that corresponds to that linear approximation being equal to zero.

So this is the analytic part, the calculations. This is the picture. Again, you're looking for the point where the linear approximation hits the x-axis. And that could be your next iteration. Your next iteration.

Is this clear? Basically, The idea behind newtons, the question that we always ask, does it always converge? No, it does not. But still, it's our favorite method to solve nonlinear equations. And you're going to see today at the very end of the class that it's the method that we actually choose if we have to solve not one nonlinear equation, but a set of nonlinear equations. Okay, so what is the problem? There are two main problems.

The first one is that for mutants to converge, and I'm gonna talk about rate of convergence after. For mutant to converge, what happens is you have to be really close to the root. And I have here one very nice example. So you have on the two graphs, You have...

the same function. So this is the function that you are working with. So this is my function.

And I want to find this 0. And let's say I start the method. So this would be one here, right?

And I start the method a little bit before the one.

So my initial value is, let's say, 0.9.

Okay, so you start here, so this would be 0.9. You take, again, as I was mentioning, you take the linear approximation. You see where the linear approximation hits the x-axis, right? So here, a little bit before the negative one. You go, you find the next linear approximation. Here is the next linear approximation. So this is the linear approximation of this function at this point. And then you see where it heats the x-axis.

And then you do the same and you can see that you are there. Right? Did you understand what, like, the picture that I was drawing? Yes.

So The idea is, let's say you start at this particular point, 0.9, a little bit before the point You start at this point, you do the linear approximation, so this could respond to the dashed line.

You hit this point. Again, you take a linear approximation of the function. You hit this point. and so on. But if you say, so the same function on the right hand side, you have the same function, but instead of starting the whole thing, instead of starting A little bit before the 1, you start a little bit after the 1. So 1.1, for example. And you do the same procedure. So you get here, right? So this is the starting point. You do the linear approximation. You get to this particular point.

You do linear approximation.

You get to the one, and you are actually not converging. You are getting away from the zero that you're trying to do. So this is a classical example where you show that a little bit of difference, so if you are A little bit. further from the root, then your method is not going to convert.

So what you say in your it depends on the initial--Yes, very much. So you have to be, Ah, sufficiently close.

to the real one. So this is one particular problem of Newton's. The other one is that it fails if you're dividing by zero. So if you're... Making the legal formula. For the updating of your E-trade if you have F prime at the particularly great if you're very close to Then it's going to be. So this is a problem. And you can imagine that if You have not a single root. But if you have a double root, So if you have a double root, that means that the first derivative, so it's not only the function that is equal to 0, but the first derivative is also equal to 0.

So this would be your double root. So if f prime is already equal to 0, you're not going to be able to make the method work. So Newton's method So this is the picture.

This is what can go wrong.

So there's a theorem that is, I'm not gonna show it. It's in the book, it takes a while, it takes about half an hour to go through the theorem. There is nothing special, I mean, not through the theorem, through the proof.

There is nothing special about it.

There's a lot of bounding. And it's interesting if you had all the time in the world, which we do not. But I'm just going to state the theorem. And I'm going to highlight the things that I would like you guys to understand. So my f has to

have a continuous f double prime in the a, b, so in the interval of interest. And you know that there is a simple root there. So let me just highlight. the relevant parts of the assumptions of the theorem.

So there's a simple root. So this theorem only works for simple roots. And then you know that there's going to be a delta, so this is a real number, a positive real number, such that starting with a given initial guess anywhere in the interval, so this is the r, the r here is the true root as before. So this is the true, well, there's no true, let me just ask it, let me just call it a root. Then the method actually converges.

quadratically but you have to start close enough.

starting with this guy in the interval that depends on the size of delta.

And usually what you have is you're going to have something that looks like this. So if you do the usual graph of iteration here and the residual here in the semi-log, you're going to see a Very nice for popular.

So again, You're working with a simple root and you have to be close enough to do certain things. So if you see that the method-So when you plot, if you see that you are not achieving a root, it's because you are not close enough to do it. Okay, so this is the MatMOM implementation. I think this is useful to look at, to play a little bit. I did it in MatLab, but of course you can do it in any other way.

Okay, so the question is, this is the same question that I did last time for the bisection method when we wrapped up last time.

Lecture.

The question is, what if we use Newton's method to find the roots of x squared equal to 0. So we are looking at Not a single word.

because it is true.

So if this is your app, Then your f prime is going to be 2 times x, which means that f prime at 0-so at the root that you're searching for is equal to 0.

So x equal to 0 is not-Uh... Well, simple mood. So what happens, so this is the case, right? Like the picture is very easy to understand. So when you, if you do the picture of the,Bravo. Well, one thing that Okay, so if I do like a parabola, like if I zoom in, See? When you're doing the updating, of newtons. So very close to the root, you're going to have a very small f.

So you have verySmall f. When you have a very So this is the updating. Updating. of your each rate K is going to be very small, so it's going to take a

longer time to converge to the root. So what happens is you lose the quadratic. Convergence rate that you have. And you have it. So when things get out of the assumptions of the theorem that I would just mention, So the theorem This theorem assures us that we're going to have quadratic convergence, Only if you are in the presencefor a simple loop and if you start close enough.

If you don't start close enough, you might not even have convergence. We just saw the example that I was showing. And many times... Even if you have convergence, the convergence, especially if you are are not in the presence of a single root.

A single root, what happens is you're going to have--A drop in the ratio. convergence in the sense that in this case of x squared, you're going to have a linear convergence of linear e. And I think there's something in the homework This is my connection. Okay. Any questions regarding news? So Newton's Any questions? Sorry. So, mutancies are a good method. But there's an effect. You have to be able to remember this is our playbook.

You have to be able to calculate the derivative of the function. So there's an extra cost. So what happens is many people start, OK, so Newton's is good because it's simple. Just do a linear approximation of the function. And instead of looking for the root of the function, you look for the root-linear approximation. That's very easy because the formula Which is this. But this is assuming that f prime You can calculate if prime and it's not, assuming it's not costly.

So some people say, "Okay, so what if Instead of working with f prime, I work with an approximation to f prime. Okay, so this is, you know,Facing. The rule. And this is what we call the secant method or the quasi-Newton's method. So the idea is, what is the motivation? This guy does not think I don't know how to calculate the derivative of my function. Or I know it, but I don't want to do it. So instead of working with f prime, I'm going to calculate the derivative of my function.

Approximation to f prime. I already have the f's. from the previous iterations.

I also know the previous iterates, so it cannot come I'm going to use an approximation that goes back to iterate k minus one. Because I know what is my iterate. I know all the iterates until... well, xk So it makes sense. I mean, it's not like too far fetched. So you go back to Newton's formula and instead of dividing By f prime, you divide by this thing. by this ratio. and you have a new updated formula.

So you have the F, so it's the same formula, the negative here. Right? Then you have f of xk, but instead of dividing by f prime, you divide by this and multiply by this. The formula is very simple. And then you don't have to calculate everything. But of course it means that you have, instead of having just one point to start, The method, as we had before, for bisection and for... What you need is you're gonna have I have to go back.

Two each ways.

You need two points to start. And it's not...

I mean, it's not what we're doing. So this is the visualization. Let's say I know I have a, oh, this is, I mean, I said it many, many times, but I'm going to say it again. You don't start looking for a root unless you know that there's one in the interval of, So we have our guys A and B, which is where I'm looking for the root. I know there's a root there. So instead of Linearizing the function, what the secant method does is instead of working with a tangent, it works with a secant.

So a secant is a line that crosses The graph of my function at two points, right? Whereas a tangent, it only touches the graph of the functions that are working at one point. So I start with two points. So A, f of A, this would be this point here. And with B, f of B, that other point.

I know everything, so I construct the tangent. So this is...

What I was showing here is like the analytical part of the formula for the secant or quasi-Newton method.

Now what we're doing when we're doing this is This is the picture. I like the picture.

I really like the picture because it reminds me, OK, so what does Newton does? I don't remember the formula. Oh, I know. It approximates with the but then I kind of see the picture and I remember what he does.

So this is the picture for sequence in the same way that I have the picture for Newton's method. Okay, so Instead of looking for where the tangent line crosses the x-axis as I did in Newton's.

I look for the point where the secant line crosses The x-axis as well. And this would be my x1. So I construct the second line. I solve for this point because I'm solving for the secant line to be equal to, for the function, for the y corresponding to the secant line to be equal to zero. And then, each standard

sequence You drop. The guy to the right-hand side. And then... You have still this point.

You calculate the other, the new point, right, corresponding to x1 and f of x1. You have another secant. So here is my secant. Oh, sorry. In this case, I dropped the A. So you keep the B. So this is something that you have to understand if you're using a library. What is your secant doing? Because some secants drop the right-hand side, and some secant methods drop the wrong side. The left hand side. And many times this dropping is completely random.

In the sense they always choose the left side Sometimes they have smart ways of deciding which one they're gonna draw. Okay, but I'm not gonna go into that. This is something that you would have to do to help. So this guy dropped the point to the left, to the left, exactly. So I kept that point and I kept this new point. I draw the second and I'm going to see where the second is. crosses the next and so forth.

Okay, yes. So now we'll move down. The second. The function is... Like we don't have the room.

No, no, no, no, this, oh, in this picture. Sure. Yeah. No, I don't have it. Just to show you that instead of looking for where the tangent crosses the X axis and in the case of the mutants, I am just doing a demonstration that in the secant case I am looking for where the graph of my secant line or where the secant line crosses the x-axis. So this is the difference between mutants and quasi-mutants.

So the iterates would be x1, x2. And now I'm going to draw-of course, I don't have the graph, right? This is what you want. But this is just a picture showing what happens when you're doing--...

Doesn't it sometimes make me divert this? Because it's going too far to the--It depends. Well, in this case, oh, you mean because it's here? No, because remember, it doesn't have to converge to do. Last time, if you guys remember, I told you that if you're doing the bisection method, sometimes it gets closer, sometimes it gets further. This also happens in these methods. So it's not-I think I told you the story, but I'm going to say it again.

So usually-If you have like an algorithm, you expect, and you're searching for something, you expect that you are, every time you iterate, you get closer and closer to the thing that you're searching. This is not true. if you're solving a nonlinear equation. Remember that last time we had like a table and I said, oh,

So this is what happens here. So you should not expect to get closer and so it's natural actually.

Oh, I think I have some pictures. Oh, I have some pictures. But now I'm showing you. You see, for example, newtons, so this is the error.

in terms of the residual, so the absolute value of F.

And this is the number of iterates. Well, I'm going to start at the beginning because now I'm just... So I'm solving for this equation, so highly nonlinear, so exponential, exponential. and I'm looking for the zeros.

In this case, I can see there are two, one here, because I just draw the graph of the function.

And I went and this is something that I think it's useful. I think maybe you're gonna do it in the homework.

What I did was I used the bisection, the secant and the newtons, and I compare, I keep the history, and I compare like the little picture that I was showing. So I have the comparison here.

So the bisection is the blue one, which I remind you has aBreak of convergence.

So linear rate of conversions.

You have newtons. that in this case, it is showing it's in the red, You can see that it has a What I wanted to show you is that you see how, for example, the bisection sometimes go up and Down, F for down. Also in the Newton, especially in the beginning, so when you're far from the root, you're going to see sometimes some oscillations in the same case as you see for the bisection. The other guy in the middle, so in the green, is the second method. And...

I think the logic of I'm going to say the rate of convergence is the golden ratio, so 1.6. So it's super linear. So it's not It's not linear, it's better, but it's not as good as quadratic as the Newton's method. So it's something Which makes sense because you are not giving. You're giving more information Because you're giving something that looks like a derivative, so you're giving more information than the bisection, but you are not actually giving the derivative, you're giving an approximation.

So the rate of convergence has to be something in between, right?

You cannot, there's, basically there's no free lunch, this is what I'm saying. Right?

So I think I say it here.

Oh, another thing is that when you're looking at the rates of convergence, I have a hard time Deciding, and I think you probably would have Deciding if the green looks like a prize. It might look like a problem if you don't know that you were working with. uh So be careful with that. The fact that he looks like a parabola doesn't mean that it's a parabola.

But I was saying that, oh, Tia.

So this is the rate of convergence of the secant method. So as with Newton's, there is no guarantee that it's going to be Conversion to something.

The initial iterates still need to be close to the group. close enough to the root It has superlinear convergence with r.

So this is not the root. This is the rate of convergence. It's 1.6. which means that it's faster than the bisection, not quite as fast as Newton's. And so we saw three Methods right by section. So you can and you saw the second method, but I cannot emphasize how many methods There are There are based on these three basic methods. Of course, if-One has to choose. Thank you. something that has to do, that is based on.

And you will see that Newton Okay, so this is what I wanted to say regarding this. Three methods. There's another method. It's called the fixed point method. or that's based on fixed point iteration. Okay, so So if you give me a function that I'm calling g here, I can define a fixed point if the fixed point exists. such as the x star, that satisfies x star is equal to the value of the function at x star.

So this means that So this is the definition of what is a fixed point. This is not the definition of the method. But the method, so this fixed point iteration is based on the definition of a fixed point.

So a fixed point-It goes like this. Okay, so if you have, let's say, Say if this is your function, Okay, and this here is a linear function. Oops. Let me just erase this one here. So we're looking at something that looks like x squared. So this is my g of x.

Okay, and this here is the line y equals to x.

So there are, if I'm looking for two fixed points of the function, I am looking for the points where the function in the x's is equal to the value of the function in the y. So this is what this means, right? So this is what it means. So you have one here. Because at this point, let's say my g of x is x squared. Then x equals to 0 would be...

The fixed point because 0 is equal to x squared equals to 0.

Zero equals. x squared evaluated at 0. So this guy here would be a fixed point.

F, the function g, so of the function, and likewise this guy would also be a fixed point because at this point the value of the function which is this one is equal which isMy wife is eating equal to the value of x.

So there are two fixed points for this equation. That's the definition there. The picture is, you look for fixed points, you graph the function, in your brain and then you look for the intersections of the graph of your function with the graph of y equals to x.

So this is the definition of a fixed point. And now comes, how can I use this definition? To solve a non-linear equation, so f of x equals to zero. So the idea is, and this is the hard part, is that you're going to use-you're going to write this equation, the equation that you're going to solve as-X of g of x. And then... So this is the idea and then the algorithm for the fixed point iteration or the fixed point iteration itself. Is that it be the next E trait?

It's gonna be the value of the function at the previous iteration. So this is the method. So let me give you an example.

Before I show you the picture, I'm going to go and show examples. The idea is So when this is solved, I have a fixed point of the function g, right? By definition, a fixed point. And The hard part is I have something to solve. This is my f of x, so this is a function that is nonlinear, and I am looking for the zeros of the function. I'm going to rewrite this guy as x equal to g of x. So I can think of, for example, g of x.

is going to be equal to x minus f of x. So when my f is equal to 0, I'm going to have g of x equals 2x.

So this would be something that I can come up with. Another function would be So I am given the F And I'm going to be looking for a G.

such that When I saw This fixed point iteration I have found the root that I'm searching for. So this is what I'm doing here. And the hard part is how do I pick the, so the whole thing about fixed point iteration is how to Come up with a good G, because not all Gs have the right properties that we would like, so that the fixed iteration works. It could also be x plus 2 times f of x, for example.

And then, well, now that I'm writing another, so these are examples here for Gs. Okay, and then... Everybody favorite.

So this is when you are...

when your fixed point iteration is actually Newton's method. Questions?

So when you say that you needTo find the G, the right properties.

Yes, let me tell you what the prop, so I gave you three Gs, right?

Three possible Gs, so this is an art, actually. It's a bit, but you have more information to decide how to go about the G. So not all G's. You can come up with the many G's that you would like, but some G's are good and some G's are not good.

This has to do with the definition of.

So, I can transform the thing that I want to solve into something that looks like this. But not all Gs are gonna suit what I need, okay? So what are the good genes, you might ask? The first is you have to make sure that the G has a fixed point. And the second part, you have to make sure that-so this is the part of existence. And then there's another part that has to do with uniqueness. And there's a theorem that guarantees The first part guarantees the existence and then it gives conditions on the G So that It exists and is unique.

So I'm going to kind of look at it.

So This part, let me just write it down.

I think I said it, but I'm going to write it down. It exists.

So the first thing that you see is that if your G is continuous, g of a is larger than a and g of b is less than or equal to b, then there exists a fixed point. Okay, so this is... something that was given and I'm not going to show you the proof this isChapter three, I think. Chapter 3 of Asher The proof of existence is based on your favorite intermediate value theorem.

So this exists. So g of a has to be larger or equal to a. And then G of B has to be-to be less than or equal to A, then there exists a fixed point in that closed region. The other thing is, so this is existence and now uniqueness. is that g prime, so the derivative of g exists, Okay? And there is A number that I'm calling rho that is less than one such that it bounds the derivative of g in the interval. And if this is so, my G is called a contraction.

So... I don't know, three things? So, oh, four things. G has to be continuous in the interval. G of A has to be larger than or equal to A, and G of B has to be less than or equal to B. to exist. Then you have to pick the G that makes this work.

So if you pick a G that looks like this. So if this is the line, y equals to x.

This is the line that crosses the 0 and the 0. If you pick something like this, then it's not going to work. Okay, so this will not work.

And Because... Well, the derivative is actually larger than the row that you have. I'm going to show you an example. So the proof of the last part is actually by contradiction. Just say suppose there are two, and then you show that you've got the contradiction. But I'm not going to go into it. If you want to look at this, you will have to look into chapter 3 of our book. And now I'm giving you the picture. The picture is actually the thing that you should remember.

Remember, I have H here. So I'm going to use an example. So I always look at x, y equals to x. So this is the graph of y equals to x. And then let me do maybe e to the-So this is the graph of E to the negative And you know there's a fixed point, here it is.

So to recognize fixed points, you draw the graph of the function. And then you see the intersection of the graph.

of the function with y equals to x. So the iteration goes like this. So you are given a point. So x Let's say it goes to 1 here. Okay, and then the I'll read them. Right here is the next one. is going to be equal to G of X of K. So to calculate-so the idea is that-Once you work through the iteration, you crack the iteration so you keep on iterating. and then eventually you hit the fixed point. So this would be the piece point here. So I start here.

So what I'm going to draw is called a cobweb. for fixed iterations. And it's the picture that everybody remembers. It goes like this.

I'm going to start at So my initial guess is x equalsClose to one. I'm going to plug my initial guess into here, and I'm going to calculate the value of the function at this point. Okay, so I put... My first step would be to come Calculate g of x0. Yes? So I'm just cracking up the machine, the algorithm, right? So I calculate this. So this would correspond-to this value here. But then what I want to do is I'm going to go and I'm going to calculate The value of the function at this point, so it's this point here, so this is going to be x1.

So this was Ersil. So I have my initial value. Thank you. I calculate the value of this x0 at the... Through the function, I get here, let me write it down.

I get G. X0. And then according to my iteration rule, I have to make it equal to x. One. So this is going to be x1 here because this is the line y equals x.

So I am at x1 here.

I do the same thing. I go to the value of the function and then I evaluate the value of the function. which would be g of x1. And then I set g of x1 equal to the value of-oh. equal to the value of this. And this would be... My X2. So it goes like this.

Here, here, here, down.

I go to this value here. I go again to the line y equals 2, x. And I just go. and do the same thing over and over so this would be my iterations so going like this until you find the value. that you're searching for. Excellent.

But for the fixed point for this to work, G prime has to be less than 1, so you have to have a contraction.

And now you can do a lot of like Interesting pictures.

For example, the one that we had was The one that we had wasSix, clear.

So this would be This is my x squared that goes like this. So let's see if I can find the points. If I start here, for example, so this would be So if my g of x is equal to x squared, Then I go, so this guy here, I start, let's say, at this point. This location, I go here. I meet the line. Here. I meet, I go, and I go to the graph of the function. Go here, then go Here I'm gonna find the zero.

I don't know, did you ever see this fixed pointCreation before. No. But it's one of the-One of the big methods If you want to find If you cannot use newtons, but you want to find the solution of the equation. And many of the methods that you've seen that I've created in the G that looks like The idea is that many of the methods that we have find a sort of secret that you then they can be transformed into a future So what sometimes we do in, let's say,White message.

Two.

Applying the numerical methods too, you will find that instead of teaching section, and you just teach fixed points. And then you do the other things as a particular type of fixed point.

But again, so... If I go...

So if I have a function that looks like this.

Let's say if I want to find this guy here. I start really close to it. Let's say I start here. So this would be my, I go to the function, I go to the line, and I'm already diverging. Because this function, Let's say... It's gonna be maybe this guy here. It does not, I mean you can play a little bit and see that it does not, comply to

these theorems. So these theorems are sufficient for you to try to find a G that has good properties.

This is what I mean by good properties. You have existence. Okay, and you have uniqueness.

But defining of the G is an art. Once you already know their words, which is when your G glues with newtons, right? or when it glues to the second membrane.

But they are not the unique genes.

They are not the only genes.

Okay, so things to remember from fixed point iteration.

The definition of what a fixed point is, The picture is You draw the graph of your function, all the points, that intersect the line y equals x.

Those are the fixed points.

And then the idea that if you want to solve f equals to f, f of x equals to zero, What you do is you come up, With the function G.

Such that, When you write 0, of the function f you are at the fixed point. So if this guy is 0, you are at the fixed point of the g. Because g of x is equal to x, and that's the definition of the fixed point. Likewise, so if this guy here is equal to 0, you are back to having g of x equals 1. But I was mentioning not all Gs. are good in the sense that they terminate the fixed point iteration. Which ones are good?

The ones that are contractions.

The derivative of the function is bounded in the interval of interest.

The humans, This two theorems, so the fixed point theorems, there's a lot of information about this. by the way, because it's such a big thing. It's like a collector of So this fixed point iteration is like a general, a way to generalize all the methods. A lot of work is done. Any questions? Okay, should I go into this? So question that you are not answering Ask, how do I know? Okay. The point exists. The point is unique. How do I know that the method converges?

The algorithm. So the fixed point iteration that I had, this is actually quite simple. It looks maybe scary,More or less scary, okay. But it's not scary. This is scarier.

Like to actually go through the proof is kind of scary. Even to go through the proof that the The mutants is quadratic. is quadratic is even worse than what

I'm going to show. Okay. I was trying-so this is the method. So this is the algorithm. And I want to show that these algorithms converge to the fixed point. I'm going to go through it really quickly because I want you guys to look at these math expressions and think, oh, this is scary.

No, sometimes you can actually understand them without a lot of... Trouble, right?

So this is what I'm going to show you with this.

So my question is...

My question is, Under the same assumptions, so the assumptions that you had before, which were My g obey is a continuous function. It has this properties and this property. So these are the assumptions that I'm concerned. width in this convergence of the fixed point.

Does the fixed point iteration converge to the fixed point?

Because it might not happen, right? And then, this is very simple. I go and I calculate the error between the fixed point, which is the star, And the kth plus one iteration. Then from here to here is just, here is the definition of the iteration process. Here is the definition of fixed point. Then this is approximated by the first order truncation with my epsilon between the previous iteration and the x star.

Is this okay everybody?

So from here to here. It's just the algorithm. From here to here is the definition of the fixed point.

Then, From here to here is scalar.

There was a series, there was expansion. So you If you have a contraction, then this G prime here is actually bounded by rho. Remember my G prime, My contraction is bounded by my rope.

So from So this here is less than or equal to rho multiplied by the magnitude of x of k minus x star.

So I have this.

If I start from the beginning, I'm going to have a lot of rows multiplying by a lot of rows, as we did before. So if I go and say, oh, but this has to be less than or equal to row power 2, and then xk minus 1 minus x hat. If you do it all the way to the first initial guess, you get rho to the power of k plus 1. But then, because rho

is smaller than what? The sequence Rho to the power k actually converges to zero, which means that you have convergence of the iteration, right?

Because if this guy is going to go to zero, the error is going to go to zero. Does it make sense? Yes, no, some yeses, some noes. No, no, no, rho is the bound on the gradient. So rho, so there exists a rho less than 1 such that the derivative We don't have gradients just yet.

The derivative is bounded by-So this is an interesting fact. Okay.

And then...

Because you have This relationship that we worked on last time This means that I have a linear rate of convergence. But my row Contrary to what we saw with the bisection is not one half.

is whatever it is depending on the G that you choose. So if you choose... a g whose derivative is very small, then the rate of convergence is going to die. It's going to be linear, but it's going to go faster, right? because your row is getting smaller and smaller by the minute. And then we can ask a question that we've asked many times before.

How many iterations do I need to do?

To reduce the error, so this error, by a factor of 10. So we've done this before. So this means that you are comparing The error at iteration k with the initial error and you are multiplying by a factor of 10.

So reducing 0.1, multiply the original error. So this is the error.

Add the duration, k, and this here is the original error. If you want to know how many case, how many iterations you have to go over, So that you drop the initial error by 10 order of magnitude, let's say, then you would have to solve the equation as we did before, and you solve just-Okay. So the number of iterations depends on the value. The upper bottom of the row. Okay, so... So we were still solving nonlinear equations.

I just want to point out, so this is like a side note.

then if you Need to calculate eigenvalues of a function.

of a matrix, if you want to calculate eigenvalues of a matrix, We have a numerical iteration, a numerical method, an iteration which is called the power method.

So this is the method. It's very, it's a very smart method. I invite you guys to look it up because it's just amazing. But... This is an iteration method. But So

the power method, what I'm saying in equation method, because the power method solves for the first eigenvalue, so the largest, the principal eigenvalue.

And then you take it away, and then you solve for the second largest, and so on. So you have to iterate in the eigenvalues.

This is what I mean. And each time you want to find one eigenvalue, you have to do an iterative process. So this is like the standard way of finding Eigenvalues, it's called the power method. But there's another way, Because if you remember, if you want to solve, so if you have an n by n matrix, Okay, so if you have an n by n matrix, what you want to do is to solve for the roots of the characteristic polynomial. You guys remember how this works?

So if you have an n by n matrix, So if you're A, A here is n by n.

Right? And you want to find the lambdas of the matrix.

You're going to do Ax equals 2 lambda x.

So the lambdas are the eigenvalues, and the x's are the eigenvectors. And then if you want to fight, so the first part If you want to find this, you solve for the determinant of A minus lambda I equals to 0.

And this here is just that, the determinant of A minus lambda I equals to zero, where here I should have a one, But what I was trying to say with this is that If you want to find the eigenvalues of a matrix, you have to find the zeros of a polynomial. And that polynomial is a nonlinear equation. So we can solve any of the methods that we, we can solve this problem with any of the methods that we have been discussing.

Instead of doing an iterative process in each one of the lambdas that you want to find, you just solve them all at the same time. that I have here, for example, One example for a four by four eight. The characteristic polynomial for this A is this guy.

So if I want to find the lambdas, which are the roots of the equation, what I need to do is, let's say, I find one interval, AB. I try to find, I bracket the interval, and I try to find the lambda that is in between that interval, just by solving mutants or by section, or it depends on how you want to proceed.

So this is One. Good thing. If you want to find, if you want to use Newton's or bisection or what have you to find the eigen the values of the matrix because you're going to find the roots of the characteristic polynomial of the matrix. See how I can kind of use this numerical stuff, numerical nonlinear equations to go ahead and find the lambdas all at the same time. Okay. So now what I'm going

to do is I want to kind of close the chapter on nonlinear equations and go into nonlinear systems of equations.

So instead of having to solve one equation, I'm going to go and solve n equations with n variables, for example. So this is my next topic. So this is solving non-linear equations in Rn.

I have n equations.

The first, so in the function f1, the second in the function f2, the nth in the function In the nth function, in the function fn, So I want to find the set of all the x's. So I want to find the vector x.

Okay, in Rn, because I have n components, that satisfy those equations being 0 all at the same time.

Is this okay? So what I'm going to do is I'm going to come up with a function f. That goes from Rn into Rm.

So I am looking to See This vector, I want to find this vector. And I have a function that I'm going to construct.

whose components are F1, F2, all the way to Fn. So for example, let's say I want to find sine of x equals to 0. Damn. cosine of x1 plus, I don't know, I'm just like coming up with this.

So plus x1 squared and then cosine of x1. x2 squared, and then cosine 1 minus x2 to the cube is equal to 0. So I'm going to come up with this function of two variables. which is sine of x1 plus x2. And another function. F2, that corresponds to the second equation. So cosine of x1 minus x2 cubed, something like that. It goes like this. So I construct the function. It is a vector value function with two components.

So this is my function f. goes from, in this case, from R2Into our two. because it's a function of two variables, the x1. the x1 and the x2 and it evaluates to a vector. So this is the vector f1 of x1, x2. And the other component is going to be F2, X1, X2. So I want to solve f is equal to 0. So this is my answer.

Is this clear? So instead of solving one nonlinear equation, I want to solve n in n variables. So the question is, how do we do it? So first of all, I think I'm going to say it, but it's obvious. It's much harder to solve a system of-Non-linear equations when you compare it to solving a system of linear equations. So this is a much harder problem. Oh, I had here an example. So this is an example for n equals to 2, but of course you can just extrapolate for whatever number of n you want. Okay. And this is what I was trying to say.

Solving nonlinear equations is harder than solving linear. So if you have a linear equation, you can actually come up and decide whether or not a solution exists just by comparing the rank of the matrix, The space where the box-the space where the functions are. But for this case, if you have a nonlinear system of equations, it's much harder to even try to determine if a solution exists.

Let alone if the solution is unique. And it was even harder to find the solution if it exists. So everything is harder. Okay.

And even if, I mean, we've seen this for the case where if you have just one variable, If you start with different initial guesses, you might go and find different routes Because you are starting closer to one root than you are to the other, so you're going to find one of the roots, not all of them. Okay. And... We thought that LU or Cholesky or QR was bad. So here in nonlinear equations, we don't even have those. Okay, so...

What did people do?

They tried to do a generalization of bisection method, of the Sicken method, and of the Newton method.

So they tried to kind of upgrade all these methods that they knew for just for solving one equation?

and make them general to solve more than one equation.

For the bisection, if I generalize the bisection, You can do it. But again, you have the same problem of slow convergence. So what people do is they go to the high order methods. And if they're going to choose, they're going to choose newtons.

And this is what I'm going to try and explain how it works. It's very simple conceptually because it's the same concept as before.

What you do is you have You kind of generate a plane or hyperplane and you try to see where the plane crosses the axis. the x-axis that you have, x1-axis, x2-axis, x3-axis, it's the same thing, only the notation gets a little bit heavier. So I'm going to kind of Um... Try to talk to your Patience, please. It's not difficult, it's just that it looks harder because the notation is heavier. So the idea is I'm going to, well, I can do it from a function that goes from Rn into R.

But I think it's easier if I just go all the way from our end to our end. So I'm just going to work on a function.

So I'm going to try to describe the Newton.

What I'm going to do is Describe Newton's method for a function that has n variables.

And it evaluates to a vector of--In our end.

So the example that I had before that I need for you guys, so this example that you have here, you have n equals to 2 because you have two variables, x1 and x2. And then you have two equations. So you have one function that I call f1 corresponding to the first equation. So my f1 is going to be sine of x1 plus x2 squared. And likewise for f2. So right now what I'm going to do is I'm going to describe Newton's method for a function that goes from Rn into Rn. whatever the n is. Bye-bye. It might look like a little bit weird, but I don't think it's going to be a problem.

So I have First, the first thing is I did with Newton's method. I'm going to come up with a linear approximation of my f function as I did before. But now my function f, instead of going from r into r, goes from rm into rm.

So each component, if you do this linear approximation, you do the linear approximation for each component of the f.

So for f1, f2, f3, all the way. So this is what you have.

This is the linear approximation of the components of my f function. So... This is the linear approximation. of my function f1 about one point z. So what's the linear approximation? This is going to be the value of the function of z plus... The gradient is the row vector. So this would be the gradient of my function f, i, evaluated at z, multiplied by x minus z.

So this x minus z is a column vector. Is this okay for everybody? Can you understand what I'm doing? So I'm just doing a tailoring expansion up to the first order. For a function that has more than one period. And I do it for, so if I had my example, what was it? I had f1 equals to sine of x1 plus x2 squared, right?

So if I do the gradient of my function here, f1, I'm going to have f1 partial with respect to x1.

And f1 partial with respect to x2, because my f1 depends on two variables.

Okay, so in this case, I'm going to have the derivative of the sine with respect to x1, so I'm looking at this.

So it's going to be the cosine of x1, and then for the second component of the gradient of f1, I'm going to find 2x squared. Sorry, 2×2, 2 times x2. So I have the linear approximation. of each one of the components of my f function. So

this is what I have there. And then, I create the approximation of my function. from a vector that happens to be the image component, the approximation of my function, of each one of the components of my function.

So you see here, All the F hats that looks like the F I hat that you had before. So you have the first component, So you have here this first component.

Four. i equals to 1, i equals to 2, all the way to i equals to n. And then you have the Jacobian matrix, whose rows are each one of the gradients of the functions that you are working with. So this first row is going to have the gradient of f1, then the last row is going to be having the gradient of fn, and each one of these gradients is multiplying the vector x1 minus the first component of z1 and so on and so forth.

So you can write this Compactly.

in this form, and this is the Jacobian matrix, which is the matrix have the first order derivatives of your f-function.

So first step, remember if you're doing Newton, the first step is linear approximation. And what I did here was to write the function The zeros I want to find as a linear approximation up to the first order. So I have This is a vector. This guy is this vector. This guy here is Jacobian matrix, is the n by n matrix.

This is the guy.

So each one of the components, so the component i, j, is the partial of fi with respect to the variable xj.

So the linear approximation is done.

Now, What did I do? What was my second step when I looked at Jutma? Instead of Setting f to 0, I would set the linear approximation to 0. Right? And solve for the... xk plus 1, imagining that the z So the center of my Taylor approximation was the x to the power k iterate. So this is exactly what I'm gonna do. So it's the same process, only the calculation looks a little bit busier.

So we speak.

So this is my approximation. I'm going to set it to 0. And I'm going to solve for The x that you see here, assuming that my x is going to be the next component. And when you do that, so if you have this equation here, when you have this, you just need to invert.

I mean, technically we know that we don't invert, right?

But you would, if you go, if you want to solve for this X, to try and determine the k plus 1, iterate, you would have to invert the Jacobian matrix.

And here you have the inversion of the matrix, or the inverse of the matrix applied on The value of the function at x. Iteration K So this here Is one way to write down Newton's method For a function that has n components, and very nice.

So, with all the reasons, While you don't converge, iterate. As follows, you calculate the Jacobian-oh, I've told you this before, but I'm going to say it again.

So once you reach, so to go from here to here, right? You just solve. It's because you, let me just write this down.

Your goal is to avoid inverting a matrix at all costs, right?

So you don't want to calculate this guy. What you do is you calculate the Jacobian and the duration x of k. You multiply by x minus-well, x plus 1-minus X of K. So this thing here, and then you set it to the negative F, and X to the power of K. So this becomes your delta x that you're going to solve for.

that you're gonna solve for using whatever method you want to solve from linear algebra. Okay, so you solve for delta x.

So this becomes Your This becomes your A. you're solving AX equals B, this becomes your X, and this minus the value of the function at the previous iteration becomes your B.

so that you avoid having to invert the digicode in the matrix. So the steps R, you compute the Jacobian, you solve. For Delta x, And then from delta x, you just calculate the next degree because you already know the previous one. So this is Newton's. So this is like a short version of Newton's algorithm if you have to solve a system of n nonlinear equations with n unknowns. So I have here an example, very simple example. Um.

So the first thing that I do is I calculate my F, which we call the residual because it's going to be the residual from the 0. So this would be how far you are from the zero that you want to calculate. You calculate the Jacobian, so you just differentiate the first function with respect to x1. With respect to x2, You differentiate here your second row. You're going to have the partials of the second function.

So this one with respect to x1 and this one again with respect to x2. And let's say you have an initial guess 1, 1. Now this is something, I mean, Do you

understand what I'm doing, basically?

Well, you don't see the calculations, but you understand how-This method, this Newton method is exactly the same in abstraction, it's doing the same thing than the method that we use for nonlinear equations, for just one linear equation. So I do...

It's just a guess. What we do in real life is if you cannot find with one guess, we keep on iterating over many guesses. Unless of course you have reason to believe that your zero is somewhere in some region, which you usually do. And you can see the convergence. Right now, I don't have, I no longer have The absolute value of f of x, but I have the norm of the vector f of x because now my f of x has f1, f2, over to fn.

So I have to calculate the norm, and you can see that it's a very nice... Quadratic convergence.

This is another example. I think this is the example in the homework. So... you have a nonlinear boundary value problem, so you have a nonlinear PDE, ODE, You have the ODE. It's nonlinear because you have the exponential here.

Okay. and you're trying to solve it in this interval and you have the boundary values equal to 0. So u of 0 is equal to u of 1, which is equal to 0. You discretize the equation in the interval with equal subintervals, h equal to 1 over n. So you discretize the domain where your ODE is valid into n equal subintervals of length h. So h is equal to 1 over n. So this discretization would give you This Manny, um...

Equations, again, you want to find your-U1, U12, U1, U2, U3, U4, U5, U6, U7, U8, U9, U10, U11, U12, U13, U14, U15, U16, U17, U18, U19, U20, U21, U22, U23, U24, U25, U26, U27, U28, U29, U30, U31, U32, U33, U34, U35, U36, U37, U38, U39, U40, Because you already know the boundary values.

The values of the functions at this 0, 0 or at 0, 1 are already equal to 0. So you only need to solve for the interior values of the u.

So you calculate the Jacobian. This is the Jacobian for this particular.

If you want, you can start with u of 0 equals to 0.

And then you just use mutants with this Jacobian to find the value.

of the equation that you are trying to solve.

So in this case, just to wrap up, Newton's method, this is something that we do not show here. I mean, we did not show for just one equation.

I'm not going to show for more than one equation. But you see that the convergence rate is the same.

Where you have m1 and m2, they are positive constants. And there, in the first statement, you have the error.

n iteration k plus one is less than or equal to theCos m1 multiplied by the norm square of the error at iteration k, and likewise for the residuals. And this is why we're saying that the convergence is quadratic. So in the homework you will need to show that you have just occur hooking of the errors, So if you want to use If you want to solve a system of nonlinear equations, One of the standard methods to do is to go ahead and Use newtons? You have to solve a system of linear equations.

The one that I was just showing in the delta x. So this system of equations that you see here, This is the But that's not a big problem because we have very fast linear solvers, so that's not our main issue, even for very large systems.

Um...

A problem is if you need to calculate the partials. So you need to calculate derivatives.

And that might be a problem. You can apply an automatic differentiator, but calculating something other than then the value of the function at a given point is always higher.

harder than just calculating the value of the function.

So if you have to calculate derivatives, it's always an added computation of costs. But okay, you can differentiate automatically. And as before, you might not, you go through all the trouble of putting this together and you might not have convergence of the methods. OK, so I've stolen a couple of minutes.

I apologize.

Yeah, so I will wrap up now being here. These squares, Next time we meet on, not next time we meet, but... on Sunday. And then we will start our next talk.

So for the pre-transparency column for classYeah. The exam is going to be the whole one hour and 30 minutes. Yes, yes. Yeah.

No, I said there's a teaching. I said it's on the whiteboard as an announcement.

Also about the grading of it, how does it work? Is there...

No, I mean...

I know this is what I'm going to say, but will there be something like normalization in terms of... Usually I don't...

So I've been doing this for a while. So I usually don't have to normalize. But I always have the 98%-And I always have the 52%. So usually I don't have to know. But if I have to, I will do it without the cost. Just do the problems that you have on the blackboard. And I suggest that you do them timely. like timing yourselves. Yes. Yes. And I've sent you a list of the things that I use. Okay, so I'll see you guys on Sunday.